4 Programming: The Frog Hop Game

You will learn

- how a program is made of commands
- how to pick commands to make the program you want
- how to make a program that works
- how to make changes and corrections to a program.

In this unit you will make a computer program. You will make a new computer game called the Frog Hop Game. You will use Scratch.



Class activity

Here is a picture of the Scratch screen:



Learning outcomes: Say what a program will do by looking at its commands; Create a simple program and remove errors so it runs

The different parts of the screen are:

- the code blocks: the blocks are stored here ready to use
- the script area: this is where you make a program
- the stage: this is where the sprite moves
- the menu bar: you can choose commands from here.

Point to the different parts of the screen.

Draw the Scratch screen. Write the names of the different parts of the screen.

user **event**
command **forever loop**
random location

Talk about...

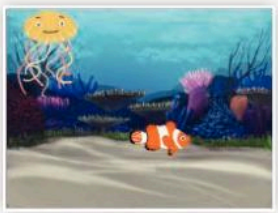
In this unit you will save your work. You must choose a file name. What makes a good file name? Talk to a classmate, then share your ideas with the rest of the class.



Did you know?

Scratch is used by children in more than 150 different countries. Scratch is available in different languages. You can make Scratch programs in more than 40 different languages.









4.1 Choose a sprite

In this lesson

You will learn:

- how to choose a sprite for your program.

Spiral back



Last year you used a simple program. It controlled a sprite. Now you will write a program to control a sprite.

In Unit 3 you played the Frog Hop Game. In this unit you will make a simple version of the game. You can choose any sprite you like. It does not have to be a frog.

You will make a Scratch program. Start on the Scratch website.

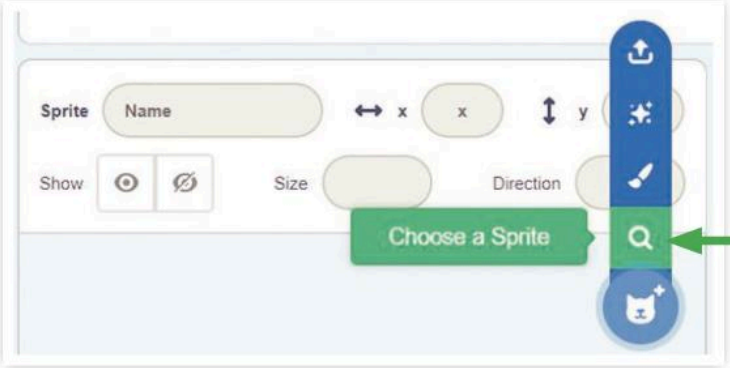
Choose a sprite

Look at the bottom right. This part of the screen shows the sprites.



Sprite 1 is a cat. Delete this sprite by clicking on the cross.

You will choose a new sprite.



Click here to see the other sprites.

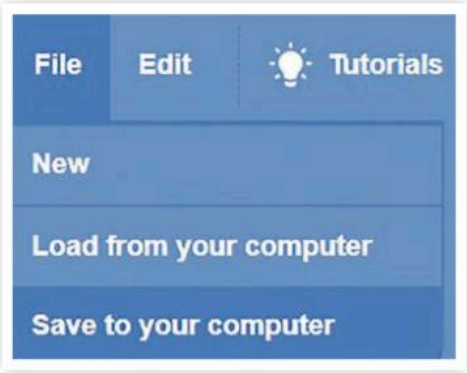
You will see pictures of sprites.



Click on the one you choose.

Save program file

Now you must save your work. You will save it on your computer. Open the File menu and choose 'Save to your computer'. Then type a name for your file.





Activity

Use the skills you learned to:

- pick a sprite
- save your file.

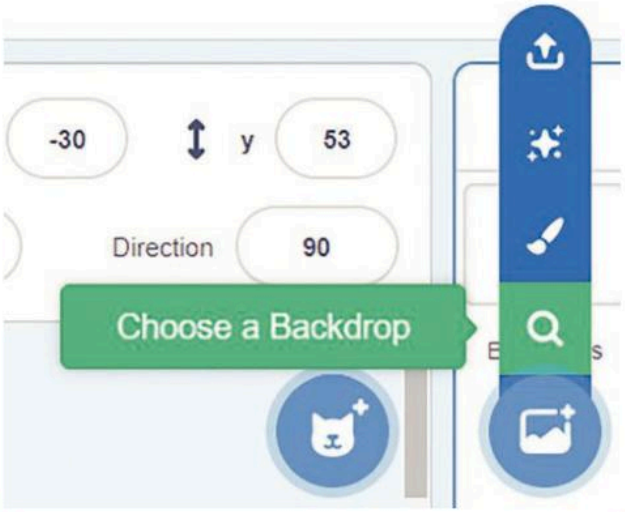


Draw the sprite you chose. Imagine a computer game with your sprite in it. Write down your idea.



Extra challenge

Add a backdrop to your program.



4.2 Command your sprite

In this lesson

You will learn:

➔ how to add commands to your program.

Load program file

Last lesson you saved your work as a file. Now you will load the file. Load means get a file from storage.



Open the file menu. Click on 'load from your computer'. Choose the file you made last time.

Coloured dots and blocks

You will make a Scratch program. A program is made of **commands**. Commands tell the computer what to do.

Scratch commands are blocks. You fit the blocks together. That makes the program.

Different types of blocks are different colours.

- 'Motion' blocks are blue.
- 'Sound' blocks are pink.

Look for the coloured dots on the left of the screen. Choose a colour to help you pick the blocks you need.

Actions

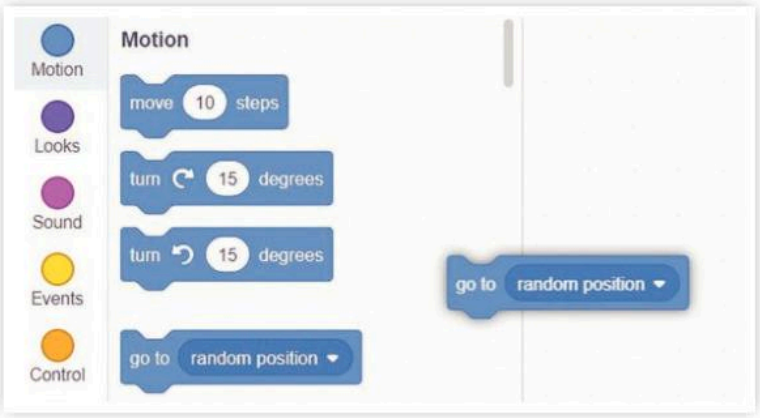
Choose a block. It will make your sprite move.

Click on the blue dot. You will see 'Motion' blocks.

Find the block that says 'go to random position'.

Drag this block to the middle of the screen.

Click on the block. The sprite will move.



Sounds

Choose a block. It will play a sound. Click on the pink dot. You will see 'Sound' blocks. Choose the 'play sound' block.

Now you have two blocks. Fit the blocks together.

Click on the two blocks. What happens?





Activity

Make the two-block program shown on this page.



Extra challenge

Add any other block to the program. See what happens when you run the program.

Think again

Say what blocks you used. Say what each block did.

4.3 A start event

In this lesson

You will learn:

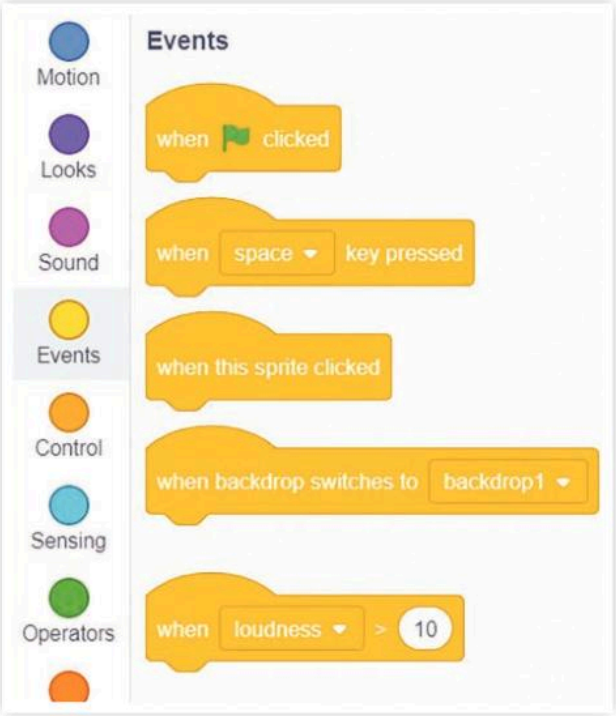
- to choose an event that starts the program.

Events

An **event** means anything that happens. In Scratch 'Events' are things the **user** does. The user is the person who uses the program. Now you will choose the 'Event' that lets the user start your program.

Click on the yellow dot that says 'Events'.

What 'Event' blocks can you see? What does each one say?

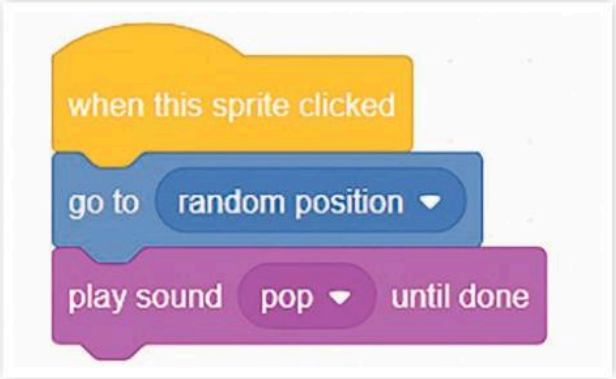


Choose an event

To choose an 'Event' drag the block to the program area.

Find the 'Event' block that says 'when this sprite clicked'.

Fit the program blocks onto the 'Event' block. Now the event will start the program.



Change colour

Now you will make the sprite change colour. Click on the purple dot that says 'Looks'. These command blocks change the look of the sprite.

Find the block that says 'change color effect by…'.

Drag this block to the script area and add it to the program.





Activity

Change your program.

- Add the 'Event' block which says 'when this sprite clicked'.
- Add the 'Looks' block which changes the colour of the sprite.

Click the sprite to make it move.

Save the file.



Extra challenge

Choose a different start event. Change the program to the new event.

Think again

Draw the 'Event' block you used. Say what it does.

If you used more than one, draw them all.

4.4 Forever loop

In this lesson

You will learn:

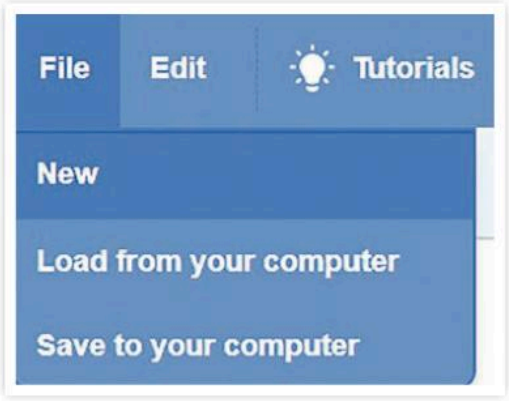
- how to change the program so the sprite moves 'forever'.

Last lesson you wrote a program. The program started when you clicked on the sprite. The sprite moved to a new place. The sprite made a sound. These things happened one time.

Now you will make a new program. In this program the sprite will move over and over again. The action will repeat.

New program

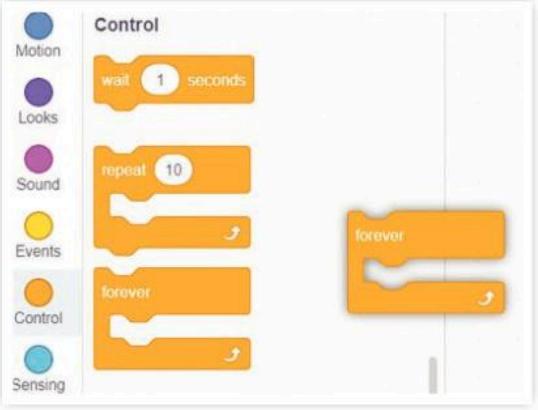
Open the file menu. Click on 'New' to make a new program.



Control blocks

Click on the orange button. You will see the orange 'Control' blocks. Find the block that says 'forever'. This block is called the **forever loop**. Drag it to the middle of the screen.

A forever loop is a way to control the program. Commands inside the forever loop will repeat.



Add commands

Here are the commands you want to repeat:

- Make the sound 'pop'.
- Go to a random position.

Put these blocks inside the forever loop. They are the same blocks you used in the last program.

Wait

Add one more block inside the forever loop. You will find it with the orange control blocks. It says 'wait 1 seconds'. This will make the sprite wait for 1 second between jumps.

The completed program looks like this.





Activity

Start a new program. Choose a sprite and a background.

Make the program shown on this page. Run the program to see what happens.

Think again

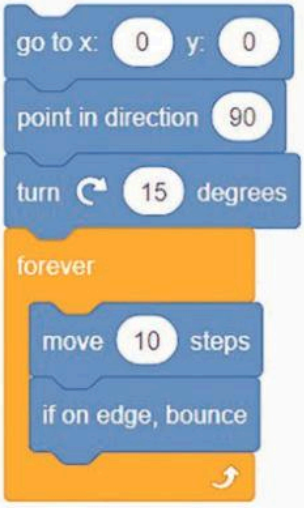
Say what happens when you put blocks inside a forever loop.



Extra challenge

Here is a new program that uses the forever loop.

- Make this program.
- Add a start 'Event' to the program.
- Change the numbers in the program and see what happens.



4.5 Plan and make

In this lesson

You will learn:

➔ how to make a new program to match a plan.

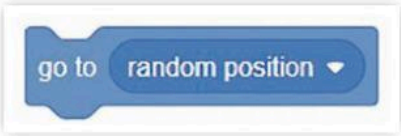
Plan

Here is the plan.

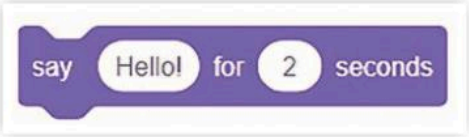
- The sprite will jump to a random place.
- The sprite will say "Hello!"
- The sprite will keep moving towards the mouse pointer.

First commands

The sprite will start in a **random** place. Random means you do not know where it will go.



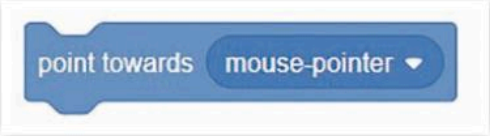
The sprite will say "Hello!"



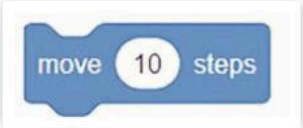
Find these blocks and fit them together.

Repeated commands

Here is the block that makes the sprite point towards the mouse pointer.



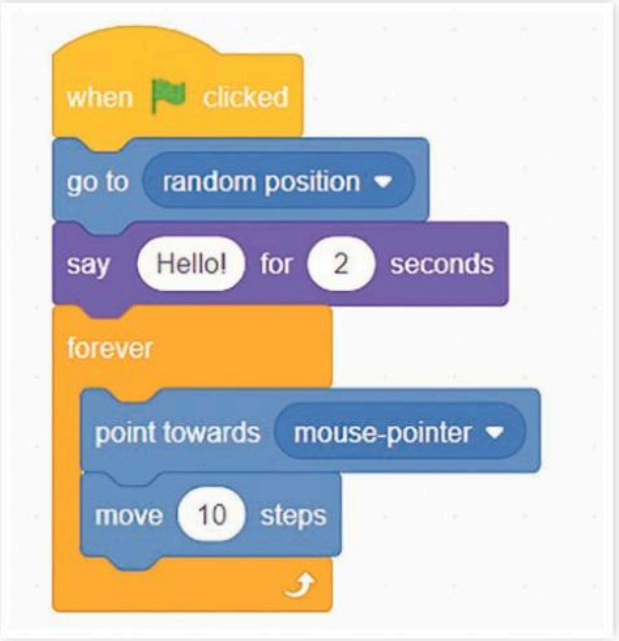
Here is the block to move 10 steps.



These two blocks will go inside a forever loop.

Completed program

Here is the program with all the pieces in place. It has a start 'Event' – the green flag.





Activity

Start a new program. Choose a sprite that looks like a fish. Choose a background that looks like underwater.

Make the program shown on this page. The program will make the fish swim about underwater.

Think again

Think of one change you could make to the program you made. What would happen if you made this change?



Extra challenge

Add a jellyfish sprite to the underwater scene. Make a program for this sprite.

- Use the same blocks as before.
- Take out the block that says 'follow the mouse pointer'.
- Add a block that says 'if on edge bounce'.
- Make the number of steps 2 instead of 10.

